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RETAINING CLIP DEVICE

Abstract

The present invention is a retaining clip device. The clip device is designed to be attached to a frame or structure. The clip device includes a hook or clip designed to be attached to a piece of equipment such as a net, tent material, or tarp. The clip device is designed to hold the piece of equipment in a position relative to the frame or structure. The clip device is adjustable on both the structure attachment and the equipment attachment such that the equipment may be situated, moved, or adjusted with respect to the frame or structure as necessary. One or more clip devices may be used to removably retain the equipment on the frame or structure in the position desired.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present invention relates to a retaining clip device. More specifically, the present invention relates to an adjustable and movable retaining clip device. Still more specifically, the present invention relates to a retaining clip device which selectively and adjustably attaches equipment to a framing structure.

2. Description of the Prior Art

[0002] There currently exists hook and clip devices for various uses designed to retain one structure to another structure. These clip devices are typically an anchoring device used with tents to attach the tent material to the pole structure. The clips are typically used with one end anchored to either the tent material or the pole structure. The second end has a hook or loop which can be attached to another receiving structure like a loop or other type of clip.

[0003] In typical formations, these clips are sewn, looped, stitched, or otherwise attached to the tent material. When utilizing the clips, they may be attached to the pole structure after the pole structure has been set up. The design is to provide a method where the tent can be broken down and stored in a compact method. When in use, the clips provide tension to erect the tent material around the pole structure.

[0004] Current clips are designed to be spaced appropriately and affixed such that they provide complimentary tension throughout the tent material to properly align the tent material and the pole structure to independently keep the tent erected through tension. The clips must be affixed to the tent material and retained in certain areas and with certain angular tension throughout the assembly to achieve the desired effect.

[0005] The clips will be sewn, stitched, or clasped onto either the tent material or the tent poles. When the user erects the tent, the clips, which are adhered to either of the structures, are then attached to the other structure. The clips are designed to be attached to a certain area to hold the tent material in a certain place and shape. If the desired shape is not adhered to, the tent material will not be held up properly and will be unusable. The clips are not designed to be placed, adjusted, moved, to other locations. The clips are not adjustable or movable such that the user can create varying structures with the tent.

[0006] What is needed is a retaining device which has a first adjustment method of attachment to a structure and a second adjustment method of attachment to a piece of equipment. The device should be removably attachable with both the first and second methods of attachment. The device should be able to be moved, adjusted, or otherwise manipulated, as desired by the user, with both attachment methods. Thus, the equipment can be moved, adjusted, re-shaped, or tensed in various ways and directions as desired by the user. The user should be able to make the equipment tighter or looser depending on the need.

SUMMARY OF THE INVENTION

[0007] The present invention is a retaining clip device. The clip is designed to retain equipment to a framing structure. The clip has a first attachment element and a second attachment element. The first attachment element is designed to be removably attached to the framing structure. The second attachment element is designed to be removably attached to the equipment. The clip allows a user to adjustably attach equipment to a framing structure to use in a designed purpose.

[0008] In an embodiment, the equipment used may be a netting material and the framing structure may be a rigid polymer structure, like PVC piping. In this embodiment, the first attachment element is designed to be removably engaged to the PVC piping. The clip may be slid, moved, or otherwise adjusted to be in an appropriate position. The second attachment element is designed to be removably and adjustably engaged with the netting material. This may be in a hook or loop type feature where the netting material can be hooked onto the clip, thereby securing the netting to the framing structure.

[0009] In other embodiments, the clip may have a first attachment element designed to be removably retained to a wooden framing structure, metal pipe, or other type of structure. Additionally, the clip may have a second attachment element designed to be removably retained to

a tarp, rope, strap, or other type of equipment feature.

[0010] The first attachment element is configured in a shape to be engaged with the framing structure. The shape of the first attachment element may be in a square, rectangle, oval, circle, or semi-circle configuration. The shape of the first attachment element is designed to provide as much friction on the framing structure as possible. The friction between the inner surface of the first attachment element allows the clip device to be held in the desired location on the framing structure.

[0011] The second attachment element is designed to be removably and adjustably retained to a piece of equipment. The equipment may be netting, rope, string, tent material, or other equipment. The second attachment element of the clip holds the equipment to the framing structure. The second attachment element is designed to easily receive or release the equipment through the use of a hook, latch, or other similar retaining feature. The retaining feature may be specifically designed to receive certain types of equipment. For example, in an embodiment, the second attachment element may be a hook to receive a looped string or rope. In another example, the second attachment element may be a broad, curved attachment designed to retain and hold a loop of a netting material.

[0012] It is a goal of the present invention to be used in conjunction with a framing structure and one or more pieces of equipment. The clip serves as a method of holding or retaining the piece of equipment in place, stabilized by the framing structure. The clip has a method of attachment to attach to the framing structure and a method of attachment to attach to the equipment. It is a goal of the clip to provide a method wherein the user can adjust positioning of the clip on the framing structure and adjust positioning of the clip on the piece of equipment to hold the equipment in the desired location and orientation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 is a top perspective view of a first embodiment of the retaining device of the present invention.

[0014] FIG. 2 is a top perspective view of a second embodiment of the device having a tightening feature on the first attachment element.

[0015] FIG. 3 is a top perspective view of the second embodiment of the retaining device of the present invention with a framing structure and netting equipment.

[0016] FIG. 4 is a first side perspective of the second embodiment of the retaining device having a tightening bolt engaged with the tightening feature on the first attachment element.

[0017] FIG. 5 is a second side perspective of the second embodiment of the retaining device having a tightening bolt engaged with the tightening feature on the first attachment element.

DETAILED DESCRIPTION OF THE INVENTION

[0018] A first embodiment of the retaining device **10** of the present invention is shown in FIG. 1. The device has a body **12**, a first attachment element **20**, and a second attachment element **40**. The first attachment element **20** has a first end **22**, a second end **24**, an inner surface **26**, and an optional tightening component system **30** of a second embodiment shown in FIG. 2. The optional tightening component system **30** has a first side **32** and a second side **34**. The second attachment element **40** has a retaining cavity **42**, an edge **44**, and an opening **46**. The first attachment element **20** is designed to be engaged with a framing structure **60**. The second attachment element **40** is designed to be engaged with equipment **80** to be releasably retained. The retained equipment **80** may have an attachment component **82**.

[0019] The device **10** is designed to removably retain together the framing structure **60** and the retained equipment **80**, such that the retained equipment **80** is held in the desired location with

respect to the framing structure **60**. In an embodiment, the framing structure **60** is a sturdy plastic structure, made out of polyvinyl chloride (PVC) pipe, for example, and the retained equipment **80** is a plastic net material. In such an embodiment, the device **10** is attached to the framing structure **60** by the first attachment element **20** and to the retained equipment **80** by the second attachment element **40**. The first attachment element **20** is configured to allow the user to move, adjust, reattach, or otherwise relocate the device **10** anywhere on the framing structure **60**. The second attachment element **40** is configured to allow the user to move, adjust, reattach, or otherwise relocate the retained equipment **80** with respect to the framing structure **60**.

[0020] The first attachment element **20** is configured to align to or complement the framing structure **60**. In an embodiment, the framing structure **60** is a circular PVC pipe. In this configuration, the first attachment element **20** is formed as a circle or semi-circle shape such that the second attachment element **40** may fit snugly around the PVC pipe **60**. In another embodiment, the framing structure **60** may be a square or rectangle piece of wood. In this configuration, the first attachment element **20** is formed in a square or rectangle shape such that the first attachment element **20** may fit snugly around the wood **60**. The material of the device **10** is selected so that the first attachment element **20** may establish at least a frictional fit about the framing structure **60**. For example, at least a portion of the device **10** may be made of a nonmetallic material, such as a plastic material.

[0021] The second attachment element **40** is configured to receive a component of the retained equipment **80**. As noted, the retained equipment **80** may be a net material, tent material, tarp, or other equipment which may be secured to the framing structure **60** by the device **10**. The second attachment element **40** is configured such that the retained equipment **80** may be placed around the second attachment element **40** and thereby retained in place. The second attachment element **40** may be formed as a hook, an extended hook, a loop, or other such configuration that may receive the retained equipment **80**. In an embodiment, the retained equipment **80** is a netting material. The device **10** is attached to the framing structure **60**. A loop or loops may form the attachment component **82** of the netting material **80** and may be placed around the second attachment element **40** wherein the netting material **80** is secured to the framing structure **60** by the device **10**. The second attachment element **40** may be made of the same material or a different material as used to make the first attachment element **20** portion of the device.

[0022] FIG. 1 depicts the first embodiment of the device **10** wherein the device **10** is configured to be engaged with the PVC pipe **60** and netting material as the retained equipment **80**. The first end **22** and the second end **24** of the first attachment element **20** are formed in a semi-circle shape to allow the user to clip the device **10** to the PVC pipe **60**. The first end **22** and the second end **24** form pressure to create a friction fit between the inner surface **26** and the PVC pipe **60** to retain the device **10** in place on the pipe **60**. The second attachment element **40** is configured to receive a loop or loops **82** of the netting material as the retained equipment **80**. The loop or loops **82** of the netting material **80** may be placed over the edge **44** into the retaining cavity **42** through the opening **46** of the second attachment element **40**. The loop or loops **82** are retained within the retaining cavity **42** to hold the netting material **80** in place. The device **10** is designed to be used with one or more of the device **10** to hold the retained equipment **80** in multiple locations along the framing structure **60**.

[0023] As depicted in FIGS. 2-5, the second embodiment of the device **10** may be configured to have an optional tightening component **30**. The tightening component **30** may be configured to have a first side **32** and a second side **34** capable of receiving a tightener **36**. The tightener **36** may be a bolt, screw, rubber band, or other method of tightening the gap between the first side **32** and the second side **34**. The tightener **36** may also have an optional nut **38** to facilitate the tightening. In this configuration, the tightening component **30** is designed to allow the user to apply additional pressure between the first end **22** and second end **24** of the first attachment element **20**. The additional pressure squeezes the first end **22** and the second end **24** of the first attachment element

20 creates additional friction between the inner surface **26** and the framing structure **60**, thereby holding the device **10** firmly in place on the framing structure **60**.

[0024] FIG. **3** depicts the device **10** in use with a PVC pipe as the framing structure **60** and the netting material as the retained equipment **80**. The inner surface **26** of the first attachment element **20** is in contact with the PVC pipe **60**, creating friction and thereby holding the device **10** in place on the pipe **60**. The retaining cavity **42** of the second attachment element **40** retains a loop **82** of the netting material **80**. The opening **46** is designed to allow the user to place or remove a loop **82** of the netting material **80** into the retaining cavity **42**. The edge **44** of the second attachment element **40** is effectively a shoulder designed to restrict the ability of the loop **82** of the netting material **80** to easily slip unintentionally off the second attachment element **40** while still making it easy to intentionally move the netting material **80** on and off the second attachment element **40**.

[0025] FIGS. **4-5** depict the device **10** with the tightener **36** engaged with the optional tightening component **30**. The tightener **36** is configured to attach to the first side **32** and the second side **34**. The optional nut **36** may be engaged with either the first side **32** or the second side **34**. As depicted in FIG. **4-5**, the tightener **36** is a hand bolt. The nut **38** is engaged with the first side **32**. The device **10** is placed around the framing structure **60**. The tightener **36**, facilitated by the nut **38**, is then twisted to bring the first side **32** and the second side **34** closer together, thus securing the device **10** onto the framing structure **60**.

[0026] The second attachment element **40** of the device **10** is equipped to receive and retain various attachment components **82** of the retained equipment **80**. For example, the attachment component **82** may be made of a string, rope, loop, or other type of component that can be hooked into and retained within the retaining cavity **42** of the second attachment element **40**. The attachment component **82** may be an existing feature of the retained equipment **80**, such as a loop of a netting material. Alternatively, the attachment component **82** may be a secondary feature added to the retained equipment **80** like a stitched or sewed loop, an attached rope, strap, or other attachment method. The second attachment element **40** may also be configured as a clasp, latch, or other type of attaching method. For example, the edge **44**, rather than being a restraining shoulder as shown, may be configured to be a latch or clasp that touches or hooks onto the body **12** of the device. In this embodiment, the edge **44** may be latched, attached, or otherwise held to the body **12** to ensure the attachment component **82** remains in the retaining cavity **42**.

[0027] The device **10** is configured for both the first attachment element **20** and the second attachment element **40** to be adjustable, movable, and able to be detached and reattached, as needed. The first attachment element **20** can be adjusted, unclipped, removed, or otherwise moved from the framing structure **60** and replaced in a location as desired. Additionally, the first attachment element **20** is configured such that the device **10** may be slid or pushed along the framing structure **60** without requiring the device **10** to be fully removed to be relocated on the framing structure **60**. The retained equipment **80** may also be adjusted, unclipped, removed, or otherwise moved within the retaining cavity **42** of the second attachment element **40**. The retained equipment **80** may be adjusted, as necessary, while still retained in the second attachment element **40**, or may be removed from and reattached to the second attachment element **40** as needed.

[0028] The device **10** is configured to be of a size to allow for the use of multiple devices **10** to be used in conjunction with each other. The framing structure **60** may be of various sizes, allowing for a plurality of the devices **10** to be used to properly align the retained equipment **80** in a shape, orientation, and alignment on the framing structure **60** as desired by the user. The device **10** may be equipped with low-profile features or have a sufficiently thin body **12** to be used in corners of the framing structure **60**.

[0029] The present invention has been described with reference to specific examples and configurations. The description is not to be taken in a limiting sense but is made merely for the purpose of illustrating the general principles of the invention, since the scope of the invention is best defined by the appended claims.

Claims

1. A retaining clip device for removably retaining one or more components to a framing structure, the device comprising: a body; a first attachment element; and a second attachment element; wherein the first attachment element is equipped to be selectively, removably, and adjustably retained to the framing structure; and wherein the second attachment element is equipped to selectively and adjustably retain thereto the one or more.
 2. The device of claim 1 wherein the first attachment element is configured in a circle or semi-circle to be retained on a circular framing structure.
 3. The device of claim 1 wherein the first attachment element is configured in a rectangle to be retained on a rectangle framing structure.
 4. The device of claim 1 wherein the first attachment element is retained to the framing structure having a shape designed to create a frictional fit on said framing structure.
 5. The device of claim 1 wherein the framing structure is made of wood, plastic, metal, or composite.
 6. The device of claim 1 wherein the second attachment element is configured as a hook.
 7. The device of claim 1 wherein the second attachment element is configured to receive and retain a rope, wire, or other net-like material.
 8. The device of claim 1 wherein the one or more components comprise a net, rope, string, or wire.
 9. The device of claim 1 wherein the first attachment element includes a tightening component.
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